

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 06

Combinational Circuit

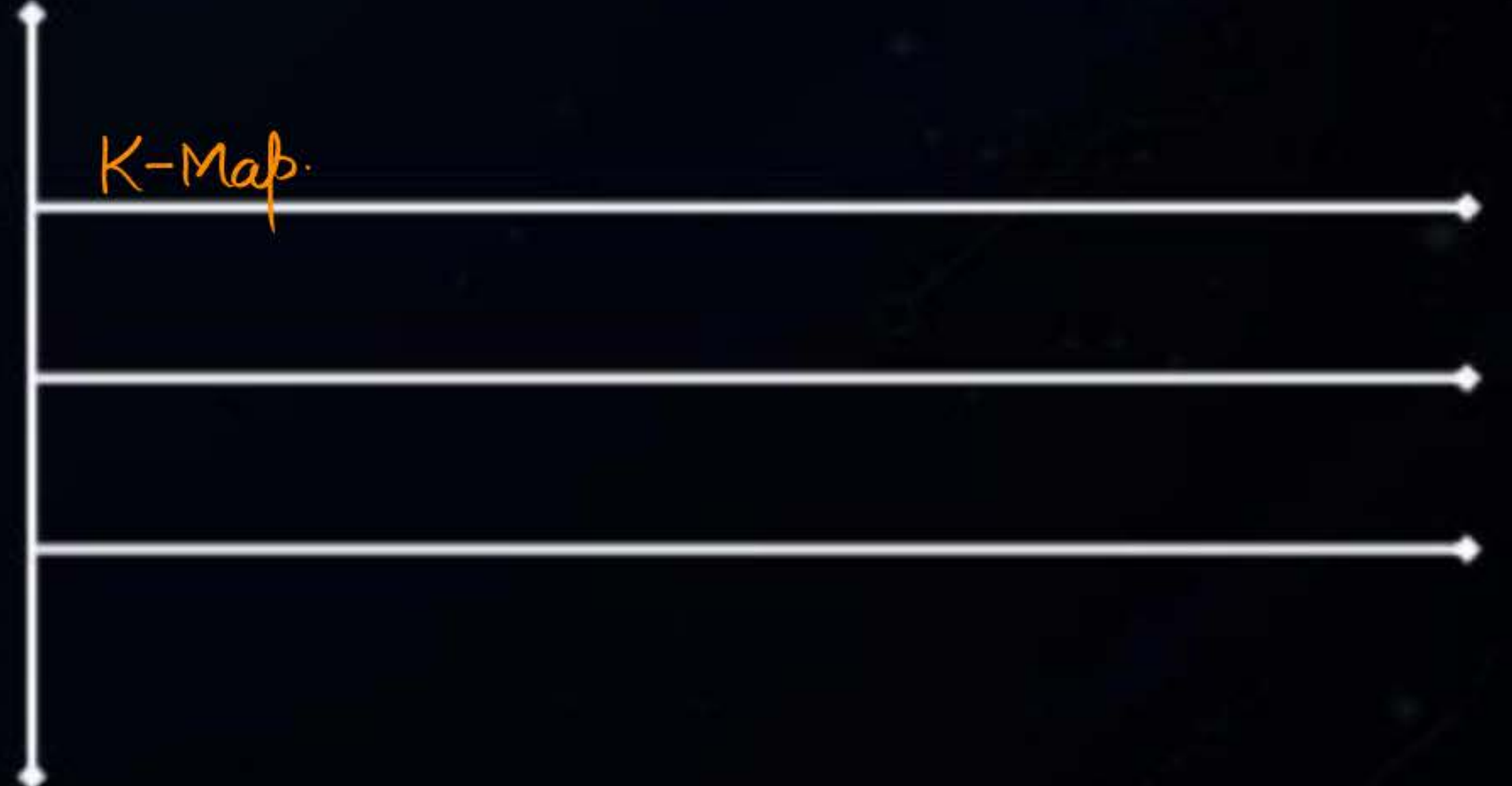
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Recap of Previous Lecture

K-Map.





Topics to be Covered

K-Map Cont.

Comparator ckt.

H.W.

Q.

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$		1	1	
$\bar{A}B$	1	1	1	1
AB	X	X		1
$A\bar{B}$	X	1	1	X

$$\bar{B}D + \bar{A}B + B\bar{D}$$

or

$$\bar{B}D + \bar{A}B + A\bar{D}$$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$		1	1	
$\bar{A}B$	1	1	1	1
AB	X	X		1
$A\bar{B}$	X	1	1	X

$$B\bar{D} + \bar{A}D + A\bar{B}$$

or

$$B\bar{D} + \bar{A}D + \bar{B}D$$

H.W.

Q.

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$			X	
$\bar{A}B$		1	X	
AB	1	1	X	1
$A\bar{B}$		1	1	

$$AB + AD + BD$$

H.W.

Q.

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$			1	
$\bar{A}B$	1		1	1
AB	1	X	X	1
$A\bar{B}$		X	1	

$$= B\bar{D} + CD$$

H.W.

Q.

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1		1
$\bar{A}B$		1	X	X
AB	1	1	X	1
$A\bar{B}$	1	1		1

$$\bar{B}\bar{D} + \bar{C}D + AB$$

or

$$\bar{B}\bar{D} + \bar{C}D + A\bar{D}$$

H.W.

Q.

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$		1	1	
$\bar{A}B$			X	
AB		X	X	
$A\bar{B}$	1	1	1	1

$$A\bar{B} + \bar{B}D$$

[Question]

A K-map is given as :

$$f(A, B, C) = \bar{A} \cdot \bar{B} \cdot \bar{C}$$

$$+ A \bar{B} \cdot 1$$

$$+ \bar{A} B \cdot 0 + A B \cdot C$$

$$f(A, B, C) = \bar{A} \bar{B} \bar{C} + A \bar{B} + A B C$$

$$f(0, 0, C) = \bar{C} + 0 + 0 = \bar{C}$$

$$f(1, 0, C) = 0 + 1 + 0 = 1$$

The solution of the K-map will be :

(a) $\bar{B} \bar{C} + AC$

(c) $\bar{A} B + B \bar{C}$

(b)

$\bar{A} \bar{B} + AC$

(d)

None of these

$$f(0, 1, C) = 0 + 0 + 0 = 0$$

$$f(1, 1, C) = C$$

	$\bar{A}$	A
$\bar{B}$	$\bar{C}$	1
B		C

$$f(A, B) = \sum(1, 2) = \bar{A} B + A \bar{B}$$

	$\bar{B}$	B
$\bar{A}$		1
A	1	

$$f(0, 0) = 0, f(0, 1) = 1$$

$$= \bar{A} \cdot \bar{B} \cdot 0 + \bar{A} B \cdot 1$$

$$+ A \bar{B} \cdot 1 + A B \cdot 0$$

$$= \bar{A} B + A \bar{B}$$

	$\bar{B} \bar{C}$	$\bar{B} C$	$B C$	$B \bar{C}$
$\bar{A}$	1			
A	1	1	1	

$$= \bar{B} \bar{C} + AC$$



	$\overline{A}$	A
$\overline{B}$	1	1
B		

$$C=0$$

or

	$\overline{A}$	A
$\overline{B}$		1
B		1

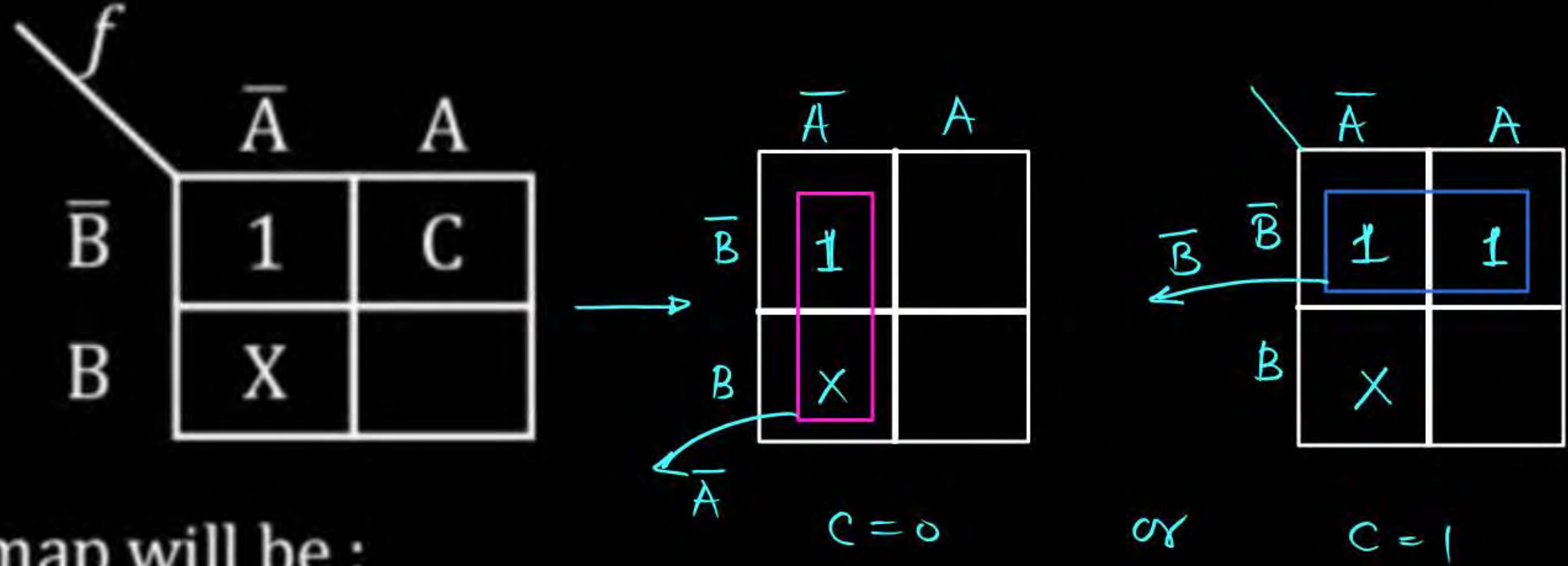
$$C=1$$

$$= \overline{C} \cdot \overline{B} + C \cdot A$$

[Question]



A K-map is given as :



The solution of the K-map will be :

(a) $\bar{A}\bar{C} + \bar{B}\bar{C}$

(b) $AC + \bar{B}\bar{C} = \bar{C} \cdot \bar{A} + C \cdot \bar{B}$

✓ (c) $\bar{A}\bar{C} + \bar{B}C$

(d) None of these

[Question]



K-map of $f(A, B, C, D)$ is given as :

$$f(A, B, C, D) = \bar{A}\bar{B} \cdot 1 + \bar{A}B\bar{D} + A\bar{B} \cdot C$$

f

	$\bar{A}$	A
$\bar{B}$	1	C
B	$\bar{D}$	

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1	1	1
$\bar{A}B$	1			1
AB				
$A\bar{B}$			1	1

$= \bar{A}\bar{B} + \bar{A}\bar{D} + \bar{B}C$

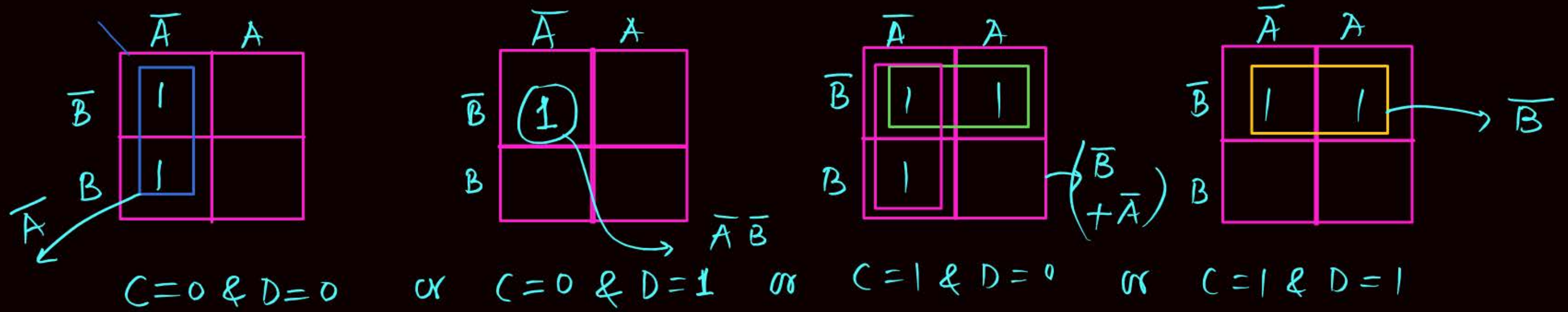
Then $f(A, B, C, D)$ will be :

(a) $\bar{A}\bar{C}\bar{D} + \bar{A}C\bar{D} + \bar{B}C\bar{D}$

(c) $\bar{A}\bar{C} + \bar{B}\bar{D} + BC$

(b) ✓ $\bar{A}\bar{B} + \bar{A}\bar{D} + \bar{B}C$

(d) $\bar{A}\bar{B} + \bar{B}\bar{D} + BC$



$$= \overline{C} \cdot \overline{D} \cdot \overline{A} + \overline{C} \cdot D \cdot \overline{A} \overline{B} + C \cdot \overline{D} \cdot (\overline{B} + \overline{A}) + C \cdot D \cdot \overline{B}$$

	$\overline{C} \overline{D}$	$\overline{C} D$	$C \overline{D}$	$C D$
$\overline{A} \overline{B}$	1	1	1	1
$\overline{A} B$	1			1
$A \overline{B}$			1	1
$A B$				

[Question]



$$f(A, B, C) = A\bar{B} + BC + \bar{A}\bar{C}$$

Then the K-map implementing above logical function will be :

(a)

f		$\bar{A}$	A
		$\bar{C}$	
$\bar{B}$			
B	1	C	

(b)

f		$\bar{A}$	A
		$\bar{C}$	1
$\bar{B}$			
B	1	C	

H.W

(c)

f		$\bar{A}$	A
		$\bar{C}$	1
$\bar{B}$			
B	1	C	

(d)

f		$\bar{A}$	A
		$\bar{C}$	1
$\bar{B}$			
B			

[**Question**] MSQ



If the function W, X, Y, Z are as :

$$W = R + \bar{P}Q + \bar{R}S$$

$$X = PQ\bar{R}\bar{S} + P\bar{Q}\bar{R}\bar{S} + \bar{P}\bar{Q}\bar{R}\bar{S}$$

$$Y = RS + \overline{PR + PQ + \bar{P}\bar{Q}}$$

$$Z = R + S + \overline{PQ + \bar{P}\bar{Q}\bar{R} + P\bar{Q}\bar{S}}$$

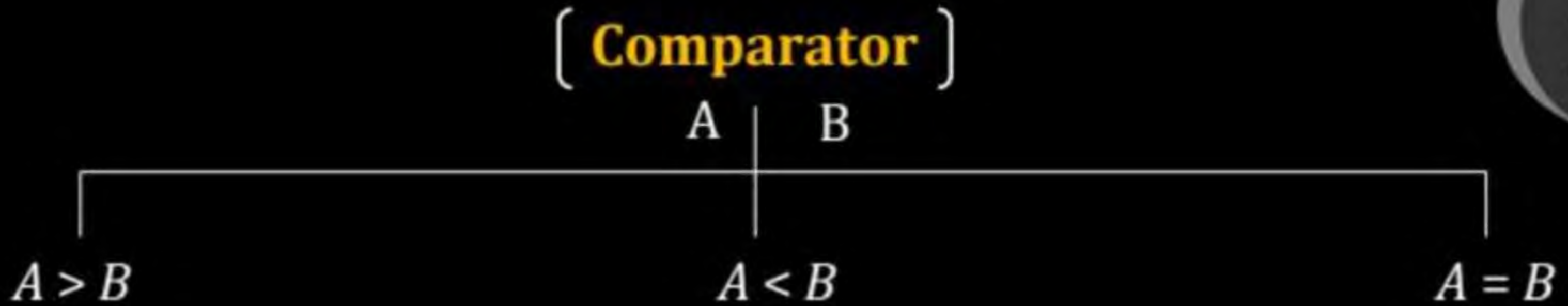
Then which of the following is/are correct ?

(a) $W = \bar{X}$

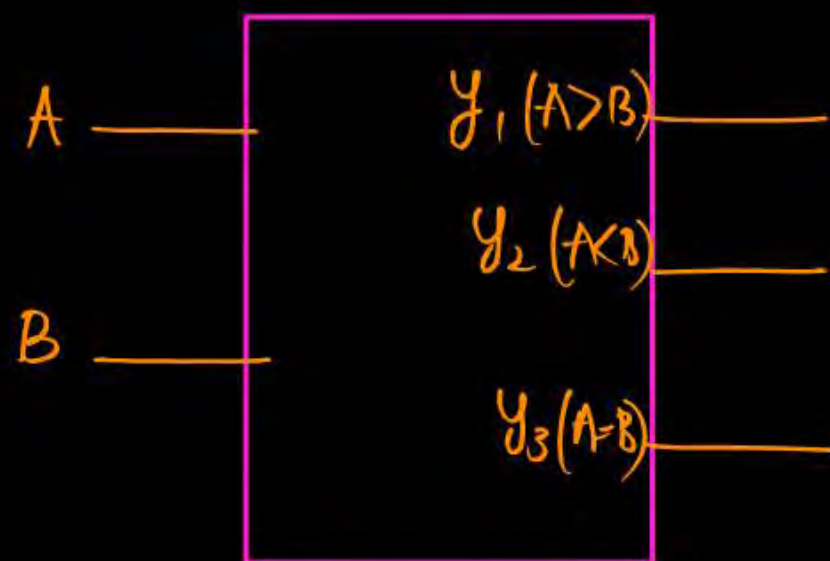
(b) $W = \bar{Z}$

(c) $\bar{W} = \bar{Z}$

(d) $\bar{X} = Z$



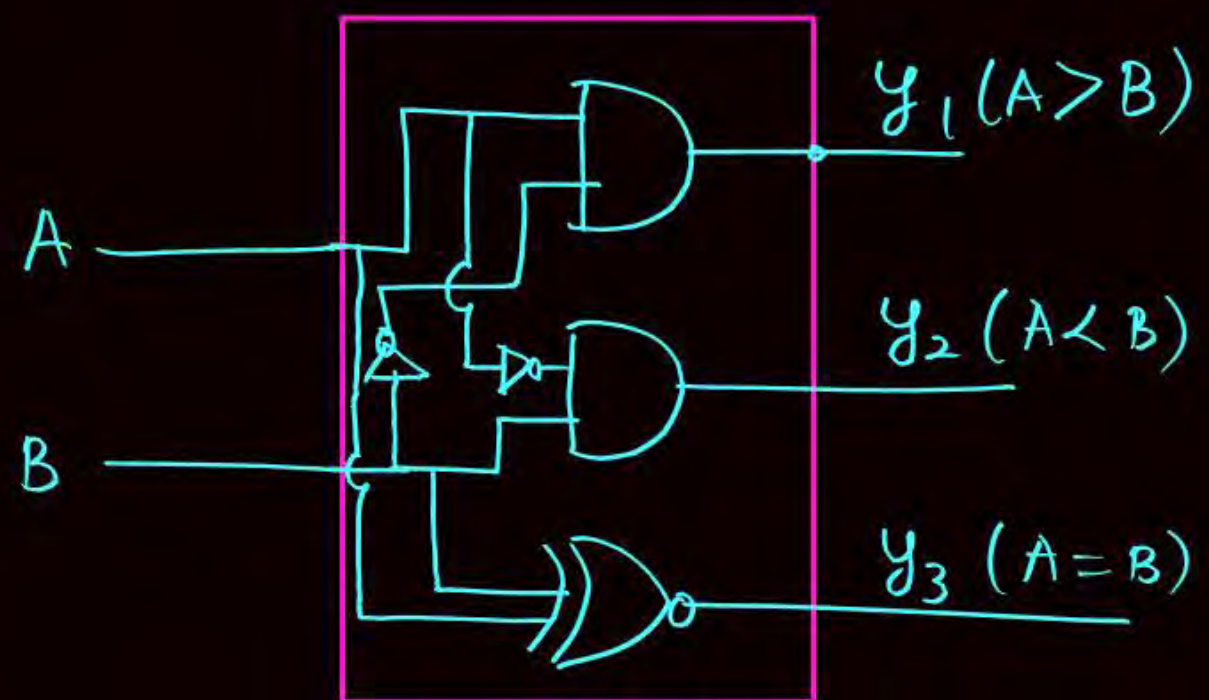
- A and B are single bit number then how to design comparator?



	A	B	$Y_1(A > B)$	$Y_2(A < B)$	$Y_3(A = B)$
0	0	0	0	0	1
1	0	1	0	1	0
2	1	0	1	0	0
3	1	1	0	0	1

$$Y_1(A, B) = \Sigma(2) = A \cdot \bar{B}, \quad Y_2(A, B) = \Sigma(1) = \bar{A} \cdot B$$

$$Y_3(A, B) = \Sigma(0, 3) = \bar{A} \bar{B} + AB = A \odot B$$



- $A = a_2 a_1 a_0$
 $B = b_2 b_1 b_0$

$$y_1(A > B) = (a_2 > b_2) \text{ or } (a_2 = b_2) \& (a_1 > b_1) \text{ or } (a_2 = b_2) \& (a_1 = b_1) \& (a_0 > b_0)$$

$$= a_2 \cdot \bar{b}_2 + (a_2 \odot b_2) \cdot (a_1 \bar{b}_1) + (a_2 \odot b_2) \cdot (a_1 \odot b_1) \cdot (a_0 \bar{b}_0)$$

$$y_2(A < B) = (a_2 < b_2) \text{ or } (a_2 = b_2) \& (a_1 < b_1) \text{ or } (a_2 = b_2) \& (a_1 = b_1) \& (a_0 < b_0)$$

$$= \bar{a}_2 \cdot b_2 + (a_2 \odot b_2) \cdot (\bar{a}_1 \cdot b_1) + (a_2 \odot b_2) \cdot (a_1 \odot b_1) \cdot (\bar{a}_0 \cdot b_0)$$

$$y_3(A = B) = (a_2 = b_2) \& (a_1 = b_1) \& (a_0 = b_0)$$

$$= (a_2 \odot b_2) \cdot (a_1 \odot b_1) \cdot (a_0 \odot b_0)$$

$$\begin{aligned} A &= a_2 a_1 a_0 \\ B &= 0 b_1 b_0 \end{aligned}$$

H.W.

$$\begin{aligned} Q. \quad A &= a_1 a_0 \\ B &= b_0 \end{aligned}$$

$$y_1 (A > B) =$$

$$y_2 (A < B) =$$

$$y_3 (A = B) =$$

$$\begin{aligned} Q. \quad A &= a_3 a_2 a_1 a_0 \\ B &= b_1 b_0 \end{aligned}$$





2 Minute Summary

- K-Map
- Comparator CKT.

Thank you

GW
Soldiers !

